Pearson Edexcel International GCSE

Mathematics A

Modular Unit 1

November 2023 Reconstructed Past Paper

STUDENT WORKBOOK

Adaptive working-space edition

31 adaptive question sections 87 marks

Selected from Linear Higher Papers 1H and 2H

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

Assistant Lecturer, Faculty of Engineering, Cairo University

B.Sc. Electrical Engineering (2017), Distinction with Honours
M.Sc. Mathematics (2023)
PhD Candidate

eliteigcse.com • WhatsApp +20 112 000 9622

About this reconstructed paper

Each question section keeps the exact source demand beside practical working space. A source grid, graph, table, or diagram is retained once at a usable size.

Ownership rule: Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. For example, drawing boundary lines supports the Unit 2 region-of-inequalities demand. This book contains the demands owned by Unit 1; the selection contains 87 marks.

This is an Elite reconstructed teaching paper selected from official Linear Higher material; it is not an official Pearson Modular examination paper. Source assets are retained for private teaching use.

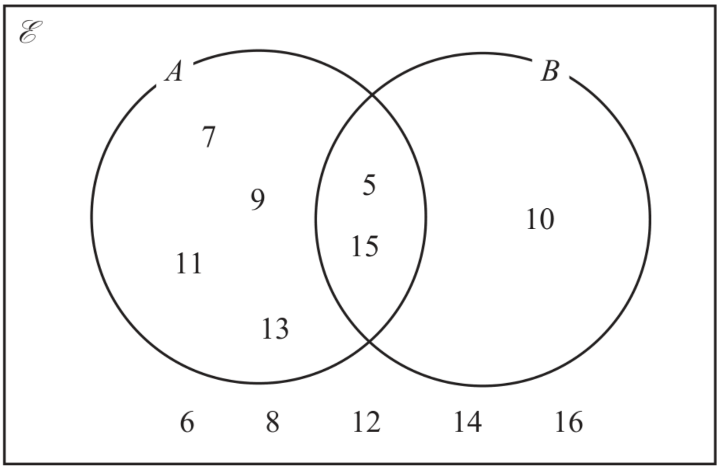
Contents

Question 01

4MA1-1H Q1 (a)

MODULAR UNIT 1
1 marks

1 Here is a Venn diagram.



List the members of the set

(a) A

....

(1)

WORKING SPACE

Working space for Question 01

Continue your method

UNIT 1
1 marks

WORKING SPACE

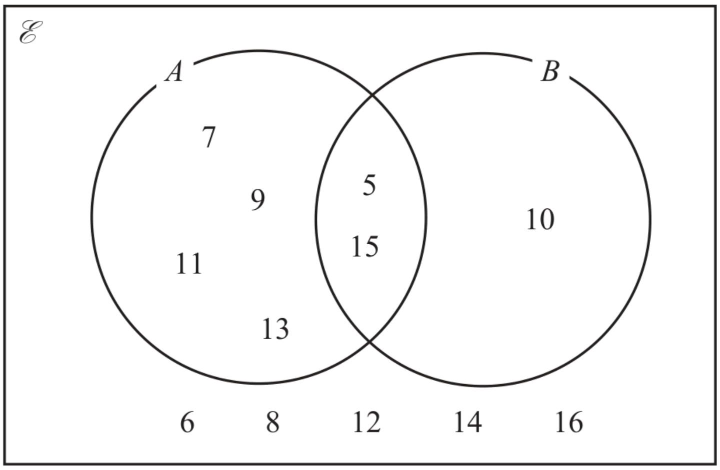
Question 02

4MA1-1H Q1 (b)

MODULAR UNIT 1

1 marks

1 Here is a Venn diagram.



List the members of the set

(b) $A \cap B$

....

(1)

WORKING SPACE

Working space for Question 02

Continue your method

UNIT 1
1 marks

WORKING SPACE

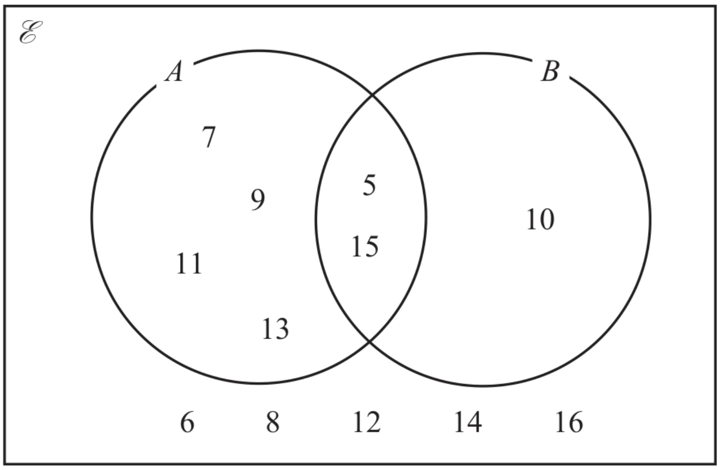
Question 03

4MA1-1H Q1 (c)

MODULAR UNIT 1

1 marks

1 Here is a Venn diagram.



List the members of the set

(c) $(A \cup B)'$

....

(1)

WORKING SPACE

Working space for Question 03

Continue your method

UNIT 1
1 marks

WORKING SPACE

Question 04
4MA1-1H Q2 (a)

MODULAR UNIT 1
2 marks

2 (a) Factorise fully $12pq - 18p$

....

(2)

WORKING SPACE

Question 05

4MA1-1H Q2 (b)

MODULAR UNIT 1
4 marks

There are 56 metal bars in a box.
Each metal bar is gold or silver or zinc.

y metal bars are gold.
$(3y + 7)$ metal bars are silver.
$(2y - 5)$ metal bars are zinc.

(b) Work out the number of metal bars that are zinc.
Show clear algebraic working.

....
(4)

WORKING SPACE

Question 06

4MA1-1H Q4 (a)

MODULAR UNIT 1
2 marks

4 A biased spinner has three sections each of a different colour.

The table shows the probability that, when the spinner is spun once, it will land on blue or on orange or on white.

Colour	blue	orange	white
Probability	0.58	$2x$	x

(a) Work out the value of x

$x =$ ....

(2)

WORKING SPACE

Question 07

4MA1-1H Q4 (b)

MODULAR UNIT 1
2 marks

4 A biased spinner has three sections each of a different colour.

The table shows the probability that, when the spinner is spun once, it will land on blue or on orange or on white.

Colour	blue	orange	white
Probability	0.58	$2x$	x

The spinner is spun 250 times.

(b) Work out an estimate for the number of times the spinner will land on blue.

....

(2)

WORKING SPACE

Question 08

4MA1-1H Q5 demand

MODULAR UNIT 1
3 marks

5 The diagram shows a shaded shape made from three identical semicircles, ***X***, ***Y*** and ***Z***

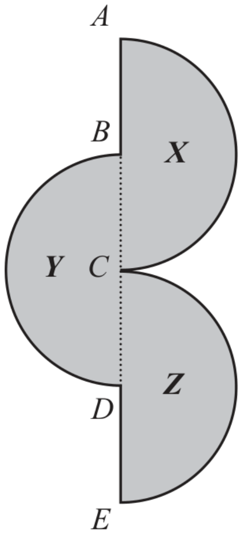


Diagram **NOT** accurately drawn

ABCDE is a straight line.

AC is the diameter of semicircle ***X*** and *B* is the centre of semicircle ***X***
BD is the diameter of semicircle ***Y*** and *C* is the centre of semicircle ***Y***
CE is the diameter of semicircle ***Z*** and *D* is the centre of semicircle ***Z***

$AC = BD = CE = 20$ cm

Work out the perimeter of the shaded shape.
Give your answer correct to the nearest whole number.

.... cm

WORKING SPACE

Working space for Question 08

Continue your method

UNIT 1
3 marks

WORKING SPACE

Question 09

4MA1-1H Q10 demand

MODULAR UNIT 1
5 marks

10 Here is triangle ABC

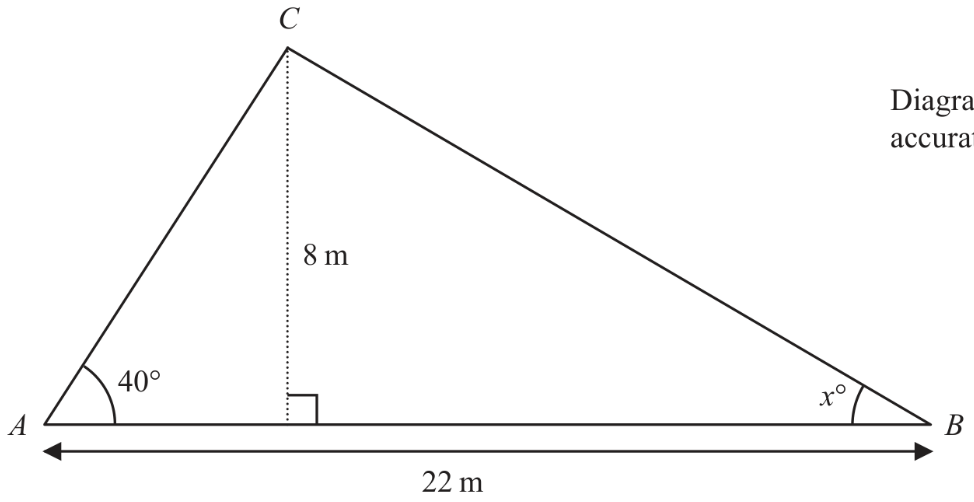


Diagram **NOT** accurately drawn

Work out the value of x
Give your answer correct to one decimal place.
Show your working clearly.

$x =$ ....

WORKING SPACE

Working space for Question 09

Continue your method

UNIT 1

5 marks

WORKING SPACE

Question 10

4MA1-1H Q11 demand

MODULAR UNIT 1

3 marks

11 Express $\frac{3}{4} + \frac{5-x}{6x}$ as a single fraction in its simplest terms.

....

WORKING SPACE

Question 11

4MA1-1H Q12 (a), (b)

MODULAR UNIT 1

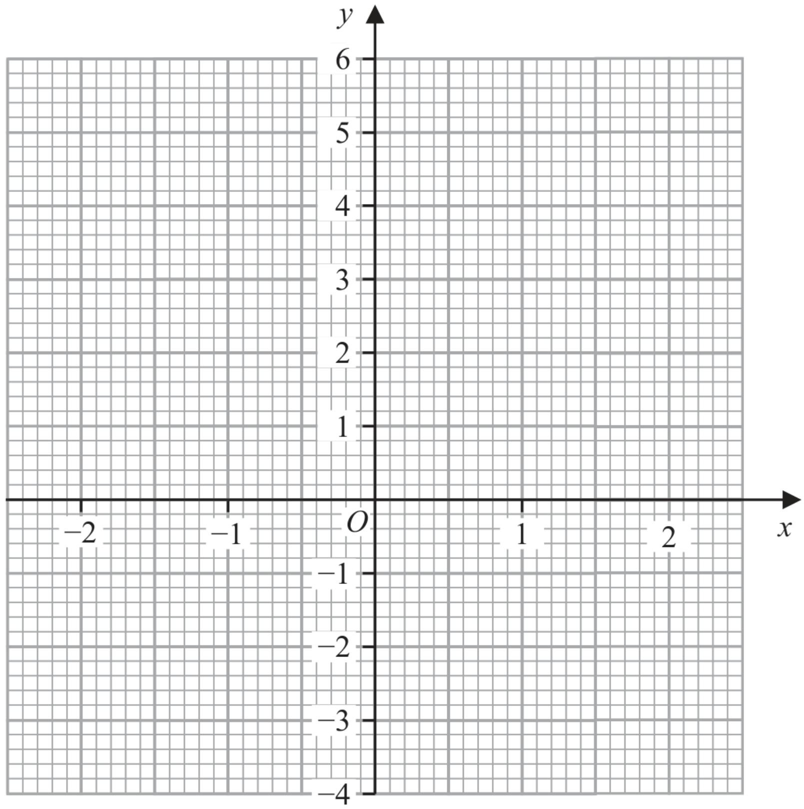
4 marks

12 (a) Complete the table of values for $y = x^3 - 3x + 1$

x	-2	-1	0	1	2
y				-1	

(2)

(b) On the grid, draw the graph of $y = x^3 - 3x + 1$ for values of x from -2 to 2

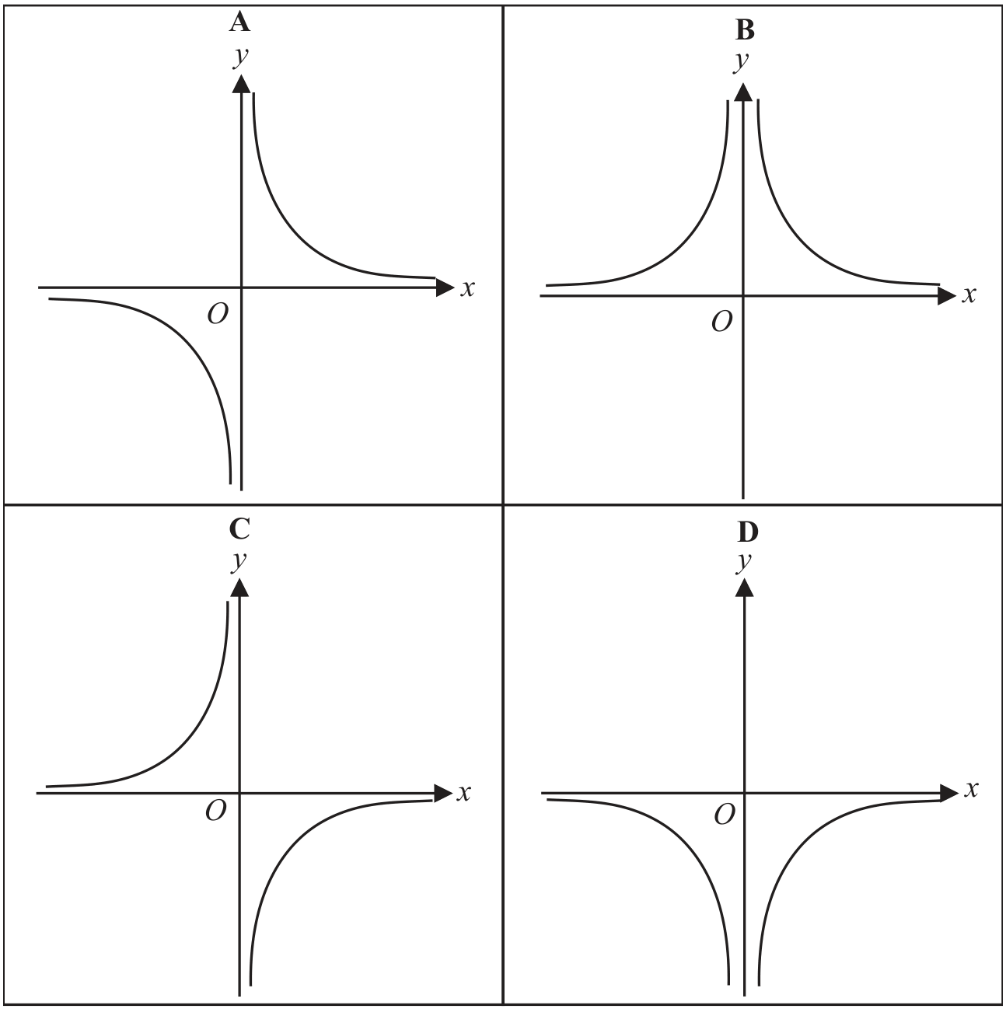


(2)

Question 12
4MA1-1H Q12 (c)

MODULAR UNIT 1
1 marks

Here are four graphs.



(c) Write down the letter of the graph that could have the equation $y = -\dfrac{1}{x^2}$

....

(1)

Working space for Question 12

Continue your method

UNIT 1
1 marks

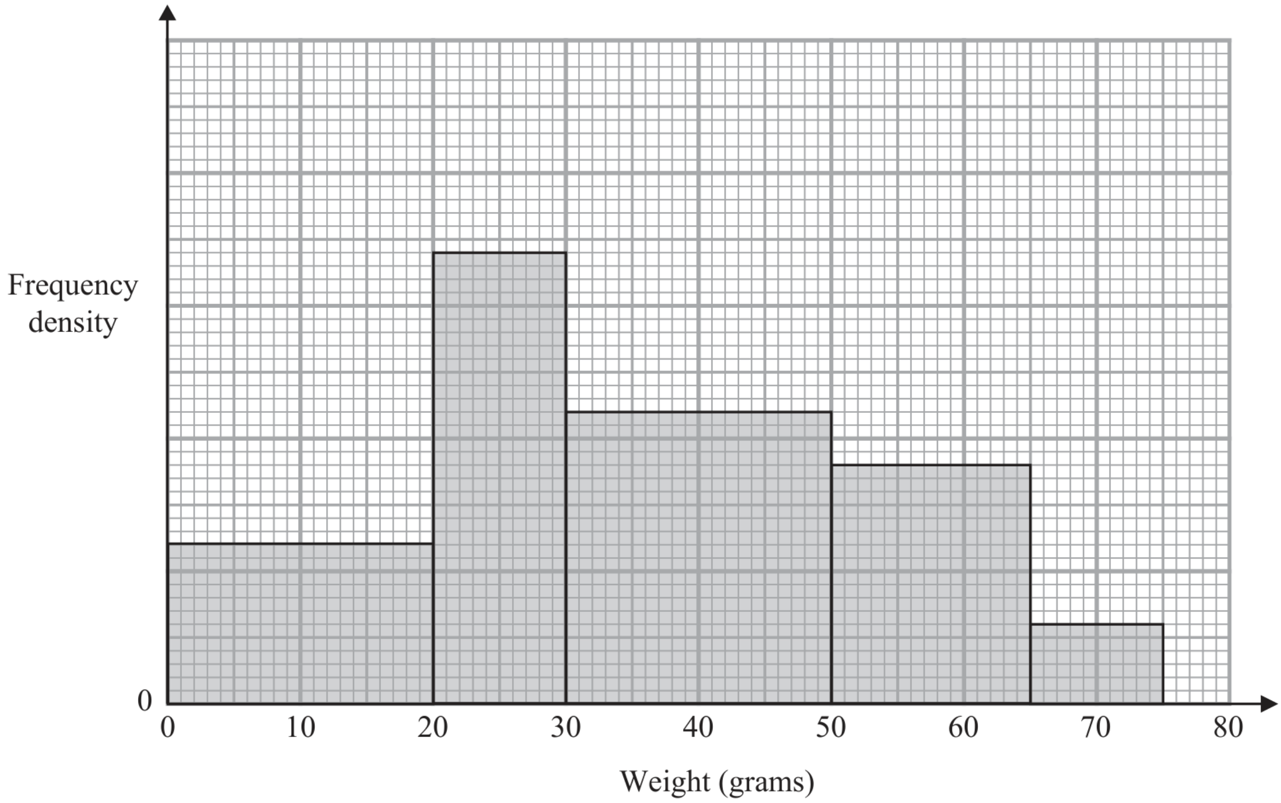
WORKING SPACE

Question 13

4MA1-1H Q17 demand

MODULAR UNIT 1
3 marks

17 The histogram gives information about the weights, in grams, of some oranges in a box.



24 of these oranges weigh less than 20 grams.

Medium oranges weigh between 35 grams and 55 grams.

Work out an estimate for the number of medium oranges in the box.

....

WORKING SPACE

Working space for Question 13

Continue your method

UNIT 1
3 marks

WORKING SPACE

Question 14

4MA1-1H Q20 demand

MODULAR UNIT 1

4 marks

20 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag. He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

....

WORKING SPACE

Question 15

4MA1-2H Q2 whole

MODULAR UNIT 1
4 marks

2 The diagram shows triangle ABC and triangle ECD

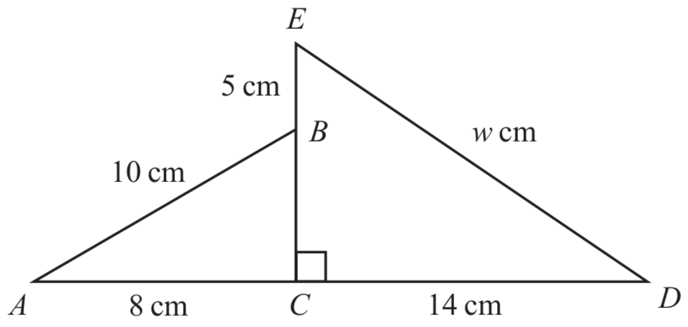


Diagram **NOT** accurately drawn

ACD and EBC are straight lines.

$AB = 10$ cm $\quad AC = 8$ cm $\quad EB = 5$ cm $\quad CD = 14$ cm $\quad ED = w$ cm

Work out the value of w
Give your answer correct to one decimal place.

$w =$ ....

WORKING SPACE

Question 16

4MA1-2H Q4 (a)

MODULAR UNIT 1
3 marks

4 (a) Solve $\dfrac{2x+5}{6} = 2x - 5$

Show clear algebraic working.

$x =$ ....

(3)

WORKING SPACE

Question 17

4MA1-2H Q4 (b)

MODULAR UNIT 1
1 marks

(b) Simplify $h^{15} \div h^{3}$

....
(1)

WORKING SPACE

Question 18
4MA1-2H Q4 (c)

MODULAR UNIT 1
2 marks

(c) Simplify fully $\left(2g^3k^5\right)^4$

....
(2)

WORKING SPACE

Question 19

4MA1-2H Q4 (d)

MODULAR UNIT 1
2 marks

(d) Given that $\dfrac{y^5 \times y^n}{y^7} = y^{12}$

work out the value of n

$n =$ ....

(2)

WORKING SPACE

Question 20

4MA1-2H Q6 whole

MODULAR UNIT 1

3 marks

6 Show that $3\frac{3}{7} \div 2\frac{2}{3} = 1\frac{2}{7}$

WORKING SPACE

Question 21

4MA1-2H Q9 whole

MODULAR UNIT 1

2 marks

9 A straight line passes through the points with coordinates (0, –3) and (2, 0)

Find an equation of the line.

....

WORKING SPACE

Question 22

4MA1-2H Q10 whole

MODULAR UNIT 1
5 marks

10 The diagram shows a hexagon *ABCDEF*

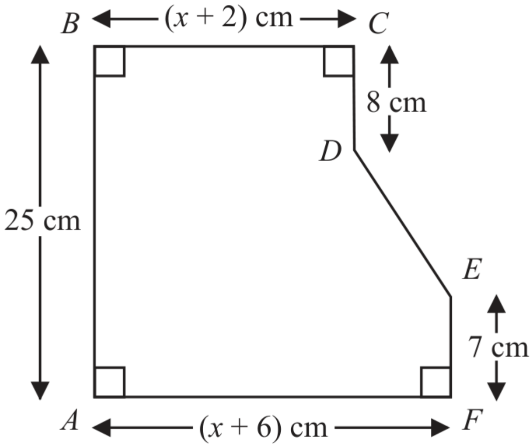


Diagram **NOT** accurately drawn

$AB = 25$ cm $\quad BC = (x + 2)$ cm $\quad CD = 8$ cm $\quad EF = 7$ cm $\quad AF = (x + 6)$ cm

The area of hexagon *ABCDEF* is 258 cm²

Work out the value of x

$x =$ ....

WORKING SPACE

Working space for Question 22

Continue your method

UNIT 1
5 marks

WORKING SPACE

Question 23

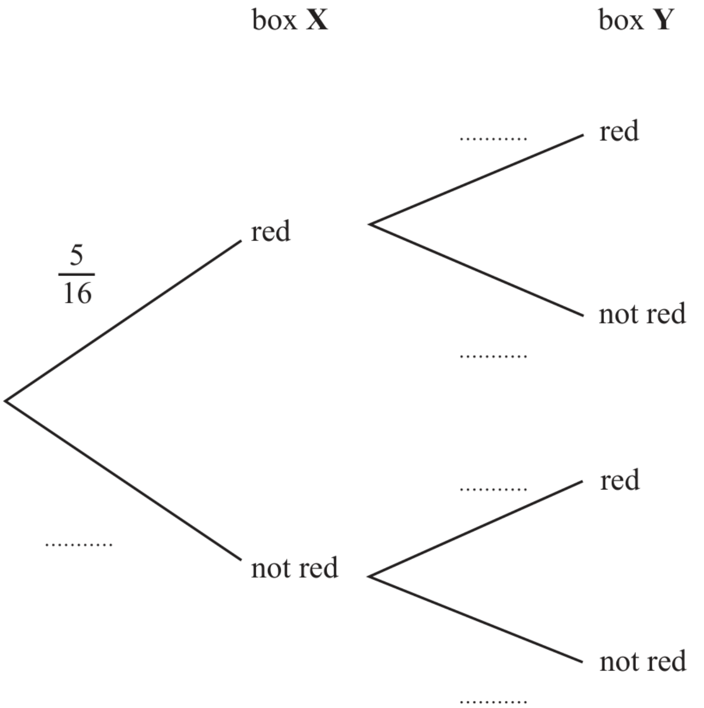
4MA1-2H Q11 (a), (b)

MODULAR UNIT 1
5 marks

11 Sid has 2 boxes of crayons, box **X** and box **Y**

5 of the 16 crayons in box **X** are red.
7 of the 20 crayons in box **Y** are red.
Sid takes at random one crayon from box **X** and one crayon from box **Y**

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Sid takes two crayons that are red or two crayons that are not red.

....

(3)

Question 24
4MA1-2H Q12 (a)

MODULAR UNIT 1
2 marks

12 $2^7 \times 4^5 = 4^x$

(a) Calculate the value of x

$x =$ ....

(2)

WORKING SPACE

Question 25

4MA1-2H Q12 (b)

MODULAR UNIT 1

2 marks

(b) Simplify fully $\left(125p^6y^{24}\right)^{\frac{2}{3}}$

....

(2)

WORKING SPACE

Question 26

4MA1-2H Q15 (a)

MODULAR UNIT 1
2 marks

15 (a) Use algebra to show that $0.3\dot{7}\dot{2} = \dfrac{41}{110}$

(2)

WORKING SPACE

Question 27
4MA1-2H Q15 (b)

MODULAR UNIT 1
3 marks

(b) Express $\dfrac{\sqrt{125}+\sqrt{80}}{\sqrt{3}}$ in the form $\sqrt{n}$ where n is an integer.

Show your working clearly.

....

(3)

WORKING SPACE

Question 28

4MA1-2H Q16 whole

MODULAR UNIT 1
3 marks

16 Expand and simplify $(2x + 3)(x - 5)(x + 4)$

....

WORKING SPACE

Question 29

4MA1-2H Q17 whole

MODULAR UNIT 1
3 marks

17 $P = a(c + y)$

$a =$ 8.3 correct to 2 significant figures
$c =$ 2 correct to 1 significant figure
$y =$ 15 correct to the nearest 5

Work out the upper bound for the value of P
Show your working clearly.

....

WORKING SPACE

Question 30

4MA1-2H Q20 whole

MODULAR UNIT 1

4 marks

20 The diagram shows equilateral triangle *ABC* with sides of length 10 cm.
A circle is drawn inside the triangle.

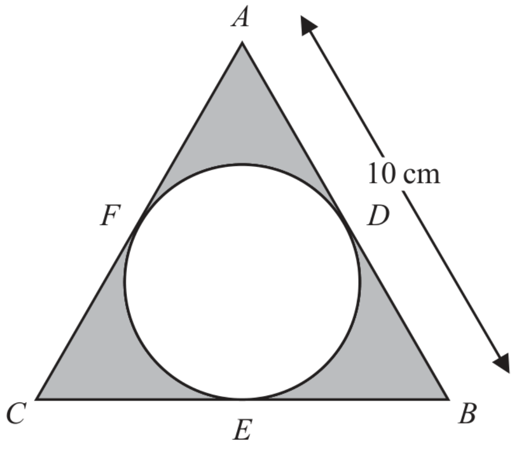


Diagram **NOT** accurately drawn

D, *E* and *F* are points on the circle.

ADB, *BEC* and *CFA* are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

.... cm^2

WORKING SPACE

Working space for Question 30

Continue your method

UNIT 1
4 marks

WORKING SPACE

Question 31

4MA1-2H Q24 whole

MODULAR UNIT 1

5 marks

24 Solve $\dfrac{45x^3 - 80x}{3x^2 + x - 4} \times \left(\dfrac{1}{3x - 4} + \dfrac{1}{x}\right) = \dfrac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x =$ ....

WORKING SPACE